Collapsing the Join Window: Soft Targets for Consolidating Fast and Slow Conversions

Anonymous

Anonymous

Abstract. Conversion pipelines are segregated by speed: a fast funnel joins on a one-day window, a slow one on a week or a month, because each must wait for its own label. We ask whether the *join* delay can be decoupled from the *attribution* window it answers to. A short join records every in-flight conversion as a negative, and most published corrections repair the record afterwards — duplicating the event, relabelling it, adding horizons — which an append-only log cannot do cheaply: under a single-visit constraint (one record per click, label frozen when written) 8 of 11 published methods violate it, up to 4.00 records per click.
What is left is a choice of target. On the public Criteo logs, across five served capacities (1,890–9,580,477 parameters) and three join delays, replacing a seven-day join with a five-minute one and letting an offline teacher stand in for the unseen conversions *improves* NE in all 15 cells, by -0.0212 to -0.0055; at a one-day window the same substitution gains -0.0096 to -0.0003. The gain grows as the funnel slows — the pipelines most worth collapsing benefit most — and the teacher may be a shipped artifact rather than a live service. The collapse is conditional: the record must keep *part* of its hard label. Discarding it entirely, the most convenient target for a consolidated pipeline, helps only at the smallest capacities and loses to waiting at the three largest.

Keywords: Conversion rate prediction · Delayed feedback · Knowledge distillation · Online learning.

1 Introduction

Conversion pipelines are segregated by speed: a purchase is joined on a one-day window, an app install or a lead on a week or a month. Each objective gets its own stream because each must wait for its own label to exist, and is stale by the width of the window it waited for. Consolidating several objectives into one model therefore meets a pipeline problem before a modelling one: K objectives with K attribution windows want K records at K times.

The obvious escape is not to wait. Write the record promptly, at a *join delay* o of minutes rather than days, and accept the label visible then. That label is systematically wrong: every conversion still in flight is recorded as a negative. A long line of work corrects the bias [1,11,10,2,3], but most corrections repair

the record after the fact — duplicating the event when its outcome arrives, relabelling it, or adding heads at further horizons — which an append-only training log cannot do cheaply. We make the restriction explicit as a *single-visit* constraint — at most one training record per click, its label frozen when written — and measure who obeys it: reproducing 11 published methods on the Criteo logs, 8 write more than one, up to 4.00 per click (Sect. 2).

What remains is one question: if the record must be written at o and carry one frozen target, what should that target be? We study four answers — from the censored outcome, through the matured one, to targets supplied by an offline teacher — across five served capacities and three join delays, and find the segregation unnecessary. Replacing a seven-day join with a five-minute one, a factor of 2016, and filling the gap with a teacher's estimate *improves* NE in all 15 cells we measure, by -0.0212 to -0.0055. At a one-day window the same substitution gains -0.0096 to -0.0003. The gain grows as the funnel slows — the opposite of the usual intuition, and it means the pipelines most worth collapsing benefit most. The teacher can be a shipped artifact rather than a live service and still beats the long-join baseline in 15 of 15 cells at seven days.

The collapse is conditional, and the condition is our second finding: the record must keep *part* of its hard label. Discarding it entirely — the most convenient target for a consolidated pipeline, carrying no attribution window at all — helps only at the smallest capacities and loses to waiting at the three largest, consistently across join delays.

2 Setting and the Single-Visit Constraint

2.1 Two delays

A click i occurs at C_i with features x_i and, if it converts, after a delay $D_i > 0$ ($D_i = \infty$ otherwise). The *attribution window* v belongs to the objective and fixes the target $Y_i^{(v)} = \mathbf{1}[D_i \le v]$, knowable only at $C_i + v$; the *join delay* o belongs to the pipeline and decides when the record is written, carrying

$$Y_i^{(o)} \;=\; \mathbf{1}[D_i \le o] \;\le\; Y_i^{(v)}. \tag{1}$$

$o = v$ is the segregated status quo; $o \ll v$ is what we want, and the inequality is the price. We write $F(o) = P(D \le o \mid D \le v)$ for the share of the target's positives visible at write time. Training runs in hourly rounds — consume the records released since the last cut-off, then score every click of the next hour once, retrospectively against $Y^{(v)}$ — so every arm is judged on the true business label whatever it trained on, and update-then-predict keeps the evaluation causal.

2.2 Two rules

Rule 1 (causality). A record's label must be computable when the record is written and must not change afterwards. Labelling a record $Y^{(v)}$ while writing

it at $C_i + o$ uses an outcome that has not happened — an upper reference (ORACLE), not a system. Strict.

Rule 2 (single-visit). With n_i the number of times click i enters the served model's loss over its lifetime, Rule 2 requires $\max_i n_i \leq 1$. Without it, shortening o is trivial — write early, write again later — so Rule 2 is what makes the question non-empty.

Two choices decide what the screen says: the criterion is $\max_i n_i$, not the mean, since it is the *existence* of a second write that obliges the pipeline to support one; and the rule governs the *served* stream, so auxiliaries are not charged. Under a stricter reading counting any auxiliary fitted on matured labels, the screen would eliminate FSIW, ULC *and both of our own arms* alike.

Reproducing 11 published methods from scratch under one protocol and counting the records each one's served model consumes per distinct click, 8 write more than one.[1] Records per click, at $v = 1\,\mathrm{d}$: DFM 1.33, ES-DFM 1.03, DEFUSE 1.03, DDFM 1.89, IF-DFM 2.79, FTP 4.00, MISS 3.05, PI 2.00 — and IF-DFM reaches 97 writes for a single click, its mechanism being an update applied whenever a label reverses. The plain re-ingestion baseline [5,4] measures 1.03 and fails the same criterion. Only FSIW, ULC, TWICE are single-visit: of these FSIW corrects the *loss* and TWICE adds a head on the conversion clock, both leaving the target untouched, while ULC alone changes the target, making it our comparator.

3 What Fills the Gap

Shortening o leaves a gap: conversions arriving between o and v are recorded as negatives, and something must stand in for them in the record's single frozen target. We study four answers, identical in pipeline, schedule, features and optimiser, differing only in the target $\tilde{y}_i$ and the age Δ at which the record is written.

FRESH ($\tilde{y}_i = Y_i^{(o)}$ at $\Delta = o$) fills the gap with nothing. WAIT ($\tilde{y}_i = Y_i^{(v)}$ at $\Delta = v$) avoids having one by deferring the write: correct label, v-old data. It is the segregated status quo and our baseline throughout — a practitioner who refuses both to duplicate records and to train on wrong labels is already running it. CORRECT-DISTILL writes at $\Delta = o$, keeping the realised label and softening only the observed negatives,

$$\tilde{y}_i \;=\; Y_i^{(o)} + (1 - Y_i^{(o)})\, w^T(x_i), \qquad w^T(x) \approx P\Big(Y^{(v)}{=}1 \,\Big|\, Y^{(o)}{=}0,\, x\Big)\,, \tag{2}$$

[1] Two measurement traps, both of which understate duplication. Dividing records consumed by records released conflates duplication with traffic variation, since a maturity-gated stream reads clicks from v ago when the click rate differed; the denominator must be *distinct clicks touched*. And a window shorter than a method's revisit horizon cannot see the revisit, so we score only clicks whose entire horizon lies inside the stream. Duplication is a property of emission logic, not of the model: re-running the audit at the smallest and largest capacities gives identical counts.

so an observed positive stays exactly 1 and only the uncertain mass moves. This is the functional form of ULC [8]; what differs is the estimator, so a comparison between them isolates that and nothing else. DISTILL-ONLY goes further and regresses on $\hat{p}_v(x) \approx P(Y^{(v)}{=}1 \mid x)$ alone: the record carries features and a number, and no observation at all.

The teacher. A separate model trained on the *matured* stream — a record enters it at $C_i + v$ carrying $Y_i^{(v)}$ — with $\hat{p}_v$ and w^T as read-outs on one trunk. It is *evaluated* at the write instant: at $C_i + o$ the student queries it and freezes the number into the record, so Rule 1 holds even though the target is soft. Softness is not an exemption from causality, only another way of satisfying it. The construction is distillation from privileged information [9]: the teacher sees the matured label, exactly what the student cannot.

Teacher capacity is fixed at 9,456,897 parameters for every student — training-only, so free at serving time, but it means *served* capacity is matched across arms while total training capacity is not. And the teacher need not be live: ours keeps learning from the matured stream, making it a service, whereas frozen after causal initialisation it is a build-time artifact. We run both. ORACLE, writing at $\Delta = o$ but labelling the record $Y_i^{(v)}$, violates Rule 1 and bounds what any correction could recover.

4 Experimental Setup

Data and protocol. The public Criteo conversion logs released with [1]. Clicks span 60 days; days $[0, 30)$ initialise the model causally and days $[30, 60)$ are the streaming period, divided into 719 hourly rounds giving 8,196,131 scored clicks at a base conversion rate of 0.1361. Learning rates are selected per arm and per cell from a 7-point grid on a held-out slice of the initialisation period, before any streaming evaluation, and pinned across seeds.

Axes. Two *attribution windows*, $v = 1\,\mathrm{d}$ and $7\,\mathrm{d}$, standing for a fast and a slow funnel; the contrast between them is the paper's point, so we never average across them and every comparison is made within one window. Three *join delays*, $o \in \{5, 30, 90\}$ minutes. Five *served capacities* from 1,890 to 9,580,477 parameters, scaling embedding tables and interaction trunk together over more than three orders of magnitude, so a result holding at one size cannot pass as general. Five seeds everywhere: 4,850 runs over 37 arms, of which this paper analyses six.

Metric. Normalised entropy against $Y^{(v)}$, lower better, pooled over all scored clicks rather than averaged over rounds — a per-round average normalises each hour by its own base rate and discards diurnal variation a model can legitimately predict. We report the difference ΔNE between two arms in one cell, since absolute NE depends on corpus and capacity.

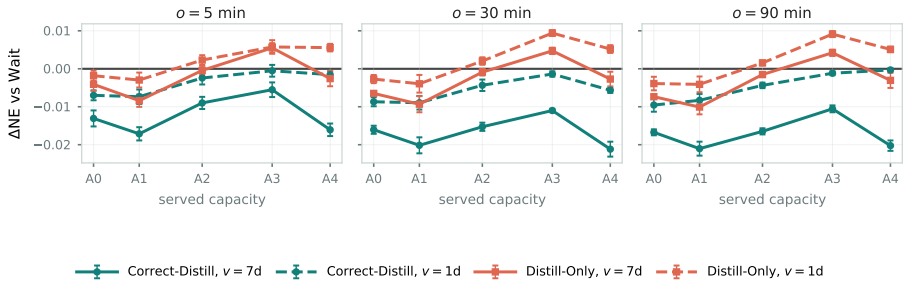


Fig. 1. ΔNE against Wait; below the line beats the segregated status quo. Colour is the target, dash the attribution window. Correct-Distill is below the line everywhere and furthest below at $v = 7\,\mathrm{d}$; Distill-Only crosses it between A1 and A2 at every join delay when $v = 1\,\mathrm{d}$. Bars are the standard deviation of the five per-seed differences.

Statistics. Seeds differ only in initialisation and minibatch order, so seed i of two arms is a matched pair: paired t-test within each cell, Holm–Bonferroni across the 15 cells of a contrast. Seed dispersion is small (median NE sd 0.00090, maximum 0.00241), so cells are individually informative. Five seeds measure initialisation variance on one data origin — precision, not generalisation.

5 Results

Throughout, ΔNE is measured against Wait — the segregated status quo that joins on the attribution window — so negative means the short join is better.

The join window can be collapsed. Writing the record 2016 times sooner than the seven-day window it is accountable to, and letting the teacher stand in for the conversions not yet seen, lowers NE in all 15 cells — every one significant — by -0.0212 to -0.0055 (Fig. 1). At the one-day window the same substitution gains -0.0096 to -0.0003, significant in 10.

The two windows differ by a factor of three in the mean (-0.0153 against -0.0048), and the direction is the useful part: *the slower the funnel, the more collapsing its join window is worth.* This inverts the intuition — a longer window censors more of the label, so the teacher has more to supply — because what dominates is the cost of the alternative: Wait at $v = 7\,\mathrm{d}$ trains on week-old data, and no label accuracy compensates.

The condition: keep part of the hard label. The collapse above used the hybrid target; the fully soft one decides how far the result can be pushed. At $v = 1\,\mathrm{d}$, Distill-Only beats Wait at the two smallest capacities (-0.0032 on average) and *loses* at the three largest (+0.0051), crossing over between A1 and A2 at every join delay (Fig. 1); all 15 cells are significant, so the crossing is not noise. A purely soft target plausibly acts as a regulariser, a variance reduction a small

model can use and a larger one no longer needs [6]. This cuts against convenience: a record with no hard label is the most portable object a consolidated pipeline could ask for, and it is exactly the one that fails at the capacities worth serving. At $v = 7\,\mathrm{d}$ the crossover is weaker and non-monotone (DISTILL-ONLY returns to -0.0028 at A4), so we state the rule for the one-day window and do not extrapolate it.

The teacher need not be live. Our teacher keeps learning from the matured stream, making it a service rather than an artifact. Freezing it after causal initialisation costs +0.00092 NE at one day and +0.00190 at seven, and frozen CORRECT-DISTILL still beats WAIT in 15 of 15 cells at seven days: a shipped model, not a running maturity-gated service, suffices.

Two anchors. Two anchors fix the scale. WAIT beats FRESH — the same short join with no correction — by -0.2066 at $o = 5$ minutes, so the gap the teacher is filling is an order of magnitude larger than the gains above, and an uncorrected short join is not an option. CORRECT-DISTILL and ULC [8] share Eq. (2) and differ only in how the corrector is estimated; they are near-indistinguishable (mean ΔNE +0.00050, 13 of 15 cells not significant at $v = 1\,\mathrm{d}$). The offline teacher does not beat the online corrector on accuracy; the case for it is architectural (Sect. 6).

6 What This Means for Consolidation

Consolidating ranking objectives into one model is an active industrial direction [7], and the obstacle is a pipeline one: under WAIT objective k's record exists only at $C_i + v_k$, so a shared append-only stream must either write K records, violating Rule 2 K times over, or delay everything to $\max_k v_k$ and impose the slowest funnel's staleness on the fastest. Our result removes the dilemma: one stream at a single short o can carry $\tilde{y}_{i,k} = Y_{i,k}^{(o)} + (1 - Y_{i,k}^{(o)})\, w_k^T(x_i)$ for every objective at once, because w_k^T is a function the pipeline calls rather than an outcome it awaits — and doing so is an improvement, most of all for the slow objectives that forced the segregation.

ULC evaluates at write time too and reaches the same accuracy, so the choice is architectural. It fits each w_k inside the served model's training loop — K objectives become K auxiliaries on one optimiser — whereas a teacher puts each w_k upstream, its only interface a number on a record. The frozen result sharpens this: that model can be a shipped artifact, so consolidation adds K build-time artifacts, not K live services.

What this does not establish. We ran one objective at a time: no evidence on interference between objectives sharing a trunk — the central risk, and what a multi-task study must measure — nor that K teachers are affordable at 9,456,897 parameters each. Our teacher is comparable in size to our largest student, so small-capacity gains partly reflect distillation from a larger model; what survives at matched capacity is the smaller effect. One corpus, one split.

Conclusion. The join window a pipeline waits for can be decoupled from the attribution window it is accountable to: shortened by three orders of magnitude, at better accuracy, with the benefit largest where the funnel is slowest — provided the record keeps part of its hard label. The fully label-free target, the most convenient to consolidate, is the one that fails. The experiment this makes worth running is K heads, K windows, one stream, teachers upstream.

References

1. Chapelle, O.: Modeling delayed feedback in display advertising. In: Proc. KDD. pp. 1097–1105 (2014)
2. Chen, Y., Jin, J., Zhao, H., Wang, P., Liu, G., Xu, J., Zheng, B.: Asymptotically unbiased estimation for delayed feedback modeling via label correction. In: Proc. WWW. pp. 369–379 (2022)
3. Ding, C., Wu, J., Yuan, Y., Li, C., Wang, X., Wang, D., Yang, F., Rabinovich, A.: Delayed feedback modeling with influence functions. Proc. AAAI **40**(17), 14648–14656 (2026)
4. Gu, S., Sheng, X.R., Fan, Y., Zhou, G., Zhu, X.: Real negatives matter: Continuous training with real negatives for delayed feedback modeling. In: Proc. KDD. pp. 2890–2898 (2021)
5. Ktena, S.I., Tejani, A., Theis, L., Myana, P.K., Dilipkumar, D., Huszar, F., Yoo, S., Shi, W.: Addressing delayed feedback for continuous training with neural networks in ctr prediction. In: Proc. RecSys. pp. 187–195 (2019)
6. Menon, et al.: A statistical perspective on distillation. In: Proc. ICML (2021)
7. Meta AI: Sequence learning: A paradigm shift for personalized ads recommendations. Meta AI Engineering Blog (2025), `https://ai.meta.com/blog/generative-ads-recommendation-model-gem/`
8. Wang, Y., Sun, P., Zhang, M., Jia, Q., Li, J., Ma, S.: Unbiased delayed feedback label correction for conversion rate prediction. In: Proc. KDD. pp. 2456–2466 (2023)
9. Xu, et al.: Privileged features distillation at Taobao recommendations (2019)
10. Yang, J.Q., Li, X., Han, S., Zhuang, T., Zhan, D.C., Zeng, X., Tong, B.: Capturing delayed feedback in conversion rate prediction via elapsed-time sampling. Proc. AAAI **35**(5), 4582–4589 (2021)
11. Yasui, S., Morishita, G., Fujita, K., Shibata, M.: A feedback shift correction in predicting conversion rates under delayed feedback. In: Proc. WWW. pp. 2740–2746 (2020)